

Physics 523 Final Exam

April, 2026.

Take home exam: You may consult a QFT textbook of your choice. You may also consult my lecture notes, your class notes, homework solutions and any handouts under the files tab at the class canvas website. No access to internet resources is allowed. Also, while working on it, please do not discuss this exam with anyone. All problems have equal weight. The exam is due by 10 am April 30. Please submit it through the canvas website. If you are writing the exam on a tablet then you may email it directly to me. If you need to contact me for questions about the exam and I am not in my office, please feel free to call at (Not after 11 pm). You may also email me to set up a meeting but it is fastest to call me.

(1) A scalar field theory described by the Lagrangian:

$$L = \frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{1}{4}\lambda\phi^4 - \frac{1}{6}g\phi^6 \quad (1)$$

- (a). What are the mass dimensions of the couplings λ and g in n space time dimensions?
- (b). Show explicitly in $n = 4$ space-time dimensions that the theory is non-renormalizable. By explicitly, I mean that you should find an example of a one-loop graph which is divergent and explain why this divergence signals non-renormalizability.

(2) Consider a quantum field theory with a fermion (field operator ψ) which also couples to a real scalar particle (field operator ϕ below). The bare lagrangian of the theory is:

$$L = \bar{\psi}(i\gamma^\mu\partial_\mu + m)\psi + \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}\mu^2\phi^2 - g\bar{\psi}\psi\phi.$$

In this problem you will consider the one loop renormalization of the theory. Consider the following rescalings to introduce the renormalization constants:

$$\psi \rightarrow \sqrt{z_2}\psi, \quad m \rightarrow \sqrt{z_m}m, \quad \phi \rightarrow \sqrt{z_3}\phi, \quad \mu \rightarrow \sqrt{z_\mu}\mu, \quad g \rightarrow z_1g$$

After this, split the lagrangian in the form, $L + L_{ct}$, where L_{ct} denotes the counter-term lagrangian.

At one loop order, calculate the divergent contribution to the renormalization counter-terms z_2, z_3, z_m, z_μ and z_1 .

(3) In lecture 22 we discussed the calculation of the process $q\bar{q} \rightarrow gg$. Please put in all the details of the calculations which were left out in the class.